Optimization for Prediction in WPM

Target-conditioned lenses, forecast aggregation and uncertainty

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29 October 2026
Mathematical methods manuscript, revised edition
DOI: 10.5281/zenodo.23782978

Abstract

This study presents predictive optimization as a sequence of target definition, causal risk estimation, compatible lens selection, uncertainty construction and decision-making. It derives minimum-risk forecast combinations, regularized adaptive weights, delayed-feedback updates, mixture distributions and a split-conformal coverage statement under exchangeability. The requested horizon remains a query rather than an unconstrained optimization variable. Worked examples distinguish forecast-error reduction from equation identification and show how cost, dependence and calibration enter a mathematically explicit WPM procedure.

Research lineage and mathematical status. This study develops a WPM mathematical extension with reference to the author's published WPM architecture [1] and foundational n/s scaling-spectrum framework [2]. The n/s paper concerns conditional moment-spectrum parameterization; the WPM paper provides the broader architectural reference. The results proved here are mathematical consequences of stated assumptions, usually applications of standard linear algebra, probability or optimization. The WPM contribution is their organization into an explicit workflow. Modeling constructions and applicability conditions are identified separately; no universal forecasting superiority is asserted.

Predictor inputs. The forecasting layer described here is data driven: it learns from observed histories and matured outcomes. It is not fed the system's true governing equation or unobserved future states at issuance. Physical equations used below define observables, explain a limiting example or specify a separately labeled comparator. Optional supplied-map inverse problems and physics-informed variants must be declared separately; they are not evidence that the default forecast knows the generator.

1 Optimize a declared predictive objective

Let $\mathcal{F}_t$ be the issuance information, T the target and P the requested horizon. A lens $r = (W, m)$ selects past extent and a declared internal resolution. The forecast can be a point, a distribution, a spatial field or an action. Its optimization objective must match that output:

$$r_T^*(t, P) \in \arg\min_r \mathbb{E}[\ell_T(Y_{t,P}, \widehat{Y}_{t,P,r}) \mid \mathcal{F}_t]. \tag{1.1}$$

The ideal expectation is unknown. An empirical matured-risk surface is an estimator, not an oracle. Optimization of that estimator can overfit, particularly when many candidates are inspected on a short history.

For squared loss the optimal point forecast is a conditional mean. For absolute loss it is a conditional median. For binary Brier loss it is a conditional event probability. The same optimal W, m need not serve all three. Positive predictive scale, an event hazard and a source coefficient are separate targets.

Proposition 1.1 (The squared-loss decomposition)**.** *Let $\mu = \mathbb{E}[Y \mid \mathcal{F}_t]$ and a be $\mathcal{F}_t$-measurable. If second moments exist,*

$$\mathbb{E}[(Y-a)^2 \mid \mathcal{F}_t] = \mathrm{Var}(Y \mid \mathcal{F}_t) + (\mu - a)^2. \tag{1.2}$$

Proof. Write $Y - a = (Y - \mu) + (\mu - a)$, expand, and use $\mathbb{E}[Y - \mu \mid \mathcal{F}_t] = 0$. □

This proves what the target is under squared loss; it does not say that a fitted lens attains it. A structurally correct equation may have larger predictive error if its estimated parameters, initial state or future inputs are uncertain.

2 Keep horizon and resolution decisions distinct

The user may request $P = 20$ steps. Selecting a more favorable $P = 5$ solves a different problem. If a decision requires several horizons, specify a horizon grid and score each at its own endpoint. An optimized warning horizon can then be a functional of coherent probabilities, not a redefinition of the original endpoint.

If a forecast for a fixed terminal time T_* is refreshed at t_k, its remaining horizon is $P_k = T_* - t_k$. Every update uses additional verified observations and is a new issuance. Compare predictions at the same cutoff; the refreshed forecast is not retrospectively an original forecast.

For a candidate bank $\mathcal{K}$ and matured origins $s + P \le t$, retain actual issued errors $e_{s,k} = \widehat{Y}_{s,P,k} - Y_{s,P}$. Historical refitting using future outcomes invalidates the issuance ledger. Loss weights can favor recent or contextually relevant origins, but their choices must be part of the model specification.

3 Fixed, Multi and Auto as distinct policies

Fixed Use a preselected eligible lens or stable reference rule. Fixed does not mean its regression coefficients can never update; it describes the lens policy.

Multi Combine several eligible same-target candidates with specified or fitted weights. Preserve useful alternatives and independent groups.

Auto Allow the lens or weights to vary with causal context, matured errors and landscape features. Regularization and a defined fallback are part of the policy.

Signed payload and richer pocket history change information representation; they do not eliminate these choices. A deeper pocket is not necessarily the best ensemble member because forecast-error dependence determines complementarity.

Let $\widehat{y}_k$ be scalar forecasts of the same target. Set $e_k = \widehat{y}_k - Y$ and $M = \mathbb{E}[ee^\top \mid \mathcal{F}_t]$. Then the risk of a linear combination with $\mathbf{1}^\top w = 1$ is

$$R(w) = \mathbb{E}[(w^\top \widehat{y} - Y)^2 \mid \mathcal{F}_t] = w^\top M w. \tag{3.1}$$

M is an error *second-moment* matrix. It equals the error covariance only when conditional biases are zero. Using centered covariance alone ignores systematic bias.

4 Optimal forecast weights and their proof trail

Theorem 4.1 (Affine minimum-risk weights)**.** *If M is positive definite, the minimizer over real weights with $\mathbf{1}^\top w = 1$ is*

$$w_* = \frac{M^{-1}\mathbf{1}}{\mathbf{1}^\top M^{-1}\mathbf{1}}, \qquad R(w_*) = \frac{1}{\mathbf{1}^\top M^{-1}\mathbf{1}}. \tag{4.1}$$

Proof. The Lagrangian is $w^\top M w - 2\lambda(\mathbf{1}^\top w - 1)$. Stationarity gives $Mw = \lambda\mathbf{1}$. The constraint sets $\lambda = (\mathbf{1}^\top M^{-1}\mathbf{1})^{-1}$. Strict convexity yields the unique global optimum, and substitution gives the risk. □

Negative weights may be unstable or violate the intended mixture semantics. Under the simplex constraint $w \geq 0$, solve a convex quadratic program rather than clipping the affine solution and assuming optimality. With a positive semidefinite singular M, a minimum exists on the compact simplex, but it may not be unique. Adding ridge or shrinkage is a decision about stability, not additional evidence.

Theorem 4.2 (Two-candidate combination)**.** *Let $A = \mathbb{E}e_1^2$, $B = \mathbb{E}e_2^2$, $C = \mathbb{E}e_1e_2$. For $\widehat{y} = (1-\alpha)\widehat{y}_1 + \alpha\widehat{y}_2$ and $D = A + B - 2C > 0$, the unrestricted optimum is $\alpha_* = (A - C)/D$. On $[0, 1]$ it is the clipped value of this scalar expression.*

Proof. Risk is $A + 2\alpha(C - A) + \alpha^2 D$. Differentiate and use convexity. If $D = 0$, then $e_1 = e_2$ almost surely and every α has the same risk. □

Worked example 4.3 (Complementarity arithmetic)**.** Take $A = 4$, $B = 9$, $C = 1$. Then $D = 11$, $\alpha_* = 3/11$ and risk is $4 - 9/11 = 35/11$. This is below both single-candidate risks because their errors are sufficiently different. If $C = 4$ with the same A, B, the constrained optimum is $\alpha = 0$: adding a candidate is not automatically useful.

Convex optimization tools for these problems are developed in [3]. No application is entitled to the oracle matrix M; it must estimate it using appropriate past forecast errors.

5 Estimating weights without inventing independence

Using normalized nonnegative historical weights a_s, estimate $\widehat{M} = \sum_s a_s e_s e_s^\top$. This matrix is positive semidefinite by construction. With many candidates and few matured origins, use $\widehat{M}_\lambda = (1-\lambda)\widehat{M} + \lambda D$, where $D \succeq 0$ is a declared shrinkage target and $0 \leq \lambda \leq 1$. If strictly positive curvature is needed, ensure the final target supplies it.

Proposition 5.1 (Duplicate invariance with mass preservation)**.** *Replacing one candidate forecast by identical copies leaves the combined forecast unchanged if their weights sum to the original weight. Treating every copy as a new equal-weight expert generally changes the forecast.*

Proof. For an identical forecast f, $\sum_j w_j f = (\sum_j w_j) f$. If equal weighting is recomputed over a larger bank, the total weight on f changes unless its group mass is explicitly preserved. □

This motivates group-normalized weights and retention of provenance. Nearby pocket cells may encode almost identical forecasts. Their count is not the number of independent opinions.

Removal utility $\widehat{\Delta}_g = \widehat{R}(w_{-g}) - \widehat{R}(w)$ depends on whether the remaining weights are refitted or merely renormalized; declare which. Positive in-sample removal utility does not prove future benefit. Temporal or run-level dependence must also remain intact when estimating uncertainty.

6 Adaptive weights and the scope of regret guarantees

A causal weight construction can minimize a regularized matured-loss objective

$$\pi_t = \arg\min_{\pi \in \Delta_K} \left\{ \sum_k \pi_k \widehat{L}_{t,k} + \eta^{-1} \sum_k \pi_k \log(\pi_k/\pi_{0,k}) \right\}. \tag{6.1}$$

For positive prior masses, Lagrange multipliers give $\pi_{t,k} \propto \pi_{0,k} e^{-\eta \widehat{L}_{t,k}}$. These are optimization weights, not posterior correctness probabilities. The reference distribution and temperature change the result.

For immediate full feedback with bounded losses $\ell_{j,k} \in [0,1]$, the standard exponential-weights method satisfies

$$\sum_{j=1}^{n} \sum_k \pi_{j,k} \ell_{j,k} - \min_k \sum_{j=1}^{n} \ell_{j,k} \le \frac{\log K}{\eta} + \frac{\eta n}{8} \tag{6.2}$$

under a uniform initial distribution. A proof uses the log ratio of successive weight sums, Hoeffding's lemma, and a lower bound from the best expert's final weight [6]. For a convex point loss, Jensen's inequality bounds the mixture forecast's loss by the weighted loss of experts. This compares with the best fixed expert on the sequence; it does not prove conditional calibration or improvement over every expert at every issuance.

With P-step delayed labels, do not apply the immediate-feedback guarantee unchanged. One simple fixed-integer-delay construction splits issuances into P residue classes and runs a separate learner in each class, whose preceding same-class outcome has matured before the next update. Summing each learner's bound gives at most $P \log K/\eta + \eta n/8$ relative to the best fixed expert, with correct event ordering. This illustrates the cost of delayed feedback. Irregular delays and partial feedback require their own analysis.

7 Anchoring a proposed correction to a reference

Let $\widehat{y}_0$ be a reference and δ a proposed correction. For $\widehat{y}_\lambda = \widehat{y}_0 + \lambda\delta$, the exact squared-risk difference is

$$R(\lambda) - R(0) = 2\lambda \mathbb{E}[e_0\delta] + \lambda^2 \mathbb{E}[\delta^2], \quad e_0 = \widehat{y}_0 - Y. \tag{7.1}$$

If $\mathbb{E}\delta^2 > 0$, the optimal unrestricted coefficient is $-\mathbb{E}[e_0\delta]/\mathbb{E}\delta^2$. A positive correction is useful only when its relation to the reference error justifies it. Statistical detection of a nonzero interaction coefficient does not establish this sign or effect on forecast loss.

Worked example 7.1 (A correction can be too large)**.** If $\mathbb{E}[e_0\delta] = -1/5$ and $\mathbb{E}\delta^2 = 1$, then $R(\lambda) - R(0) = -2\lambda/5 + \lambda^2$. The optimum is $\lambda = 1/5$ with risk reduction $1/25$. Full correction $\lambda = 1$ increases risk by $3/5$. Detecting a real signal does not justify deploying it at full estimated strength.

Setting $\lambda = 0$ preserves the reference algebraically. Choosing λ from noisy historical estimates does not guarantee noninferiority in an unknown future regime.

8 Turn a forecast equation into a distribution

One location–scale construction is

$$Y_{t,P} = \mu_{t,P} + s_{t,P} Z_{t,P}, \qquad s_{t,P} > 0. \tag{8.1}$$

The scale can be $e^{g_\eta(z_t,P)}$ to enforce positivity. Estimate standardized errors only from matured forecasts. If their proposed conditional CDF is $G_{t,P}$, the predictive CDF is $F(y) = G_{t,P}((y-\mu)/s)$.

For compatible pocket distributions F_i, use $F(y) = \sum_i \pi_i F_i(y)$ with nonnegative normalized weights. It is a valid CDF because each term is monotone, right-continuous and has the appropriate limits. Its mean and variance are

$$\mu = \sum_i \pi_i \mu_i, \qquad v = \sum_i \pi_i \{v_i + (\mu_i - \mu)^2\}. \tag{8.2}$$

Variance includes between-pocket disagreement. A mixture is different from a linear average of random variables, whose variance depends on covariance. For any interval $[a, b]$, probability is $F(b) - F(a^-)$; for a continuous law this is $F(b) - F(a)$.

Worked example 8.1 (A mixture does not have just average variance)**.** For $\frac{3}{4}N(0,1) + \frac{1}{4}N(2,4)$, the mean is $1/2$. Average within-component variance is $7/4$, and between-component variance is $3/4$. Total variance is $5/2$. Omitting disagreement understates uncertainty.

Probability normalization is an algebraic property. Reliability is a statement about forecasts and outcomes together. Proper scores are designed for that distinction [4].

9 A coverage theorem with explicit assumptions

Fit $\widehat{\mu}, \widehat{s} > 0$ using training data. On n separate calibration pairs define $R_j = |Y_j - \widehat{\mu}(X_j)|/\widehat{s}(X_j)$. Let $k = \lceil (n+1)(1-\alpha) \rceil$ and use the kth order statistic q, with $q = \infty$ if $k = n+1$.

Theorem 9.1 (Split-conformal marginal coverage)**.** *Conditional on the training data, if calibration pairs and the future pair are exchangeable and use the same frozen scoring rule, then*

$$\mathbb{P}\{Y_{n+1} \in [\widehat{\mu}(X_{n+1}) - q\widehat{s}(X_{n+1}), \widehat{\mu}(X_{n+1}) + q\widehat{s}(X_{n+1})]\} \geq 1 - \alpha. \tag{9.1}$$

Proof. With no ties, exchangeability makes the future score's rank uniform among $n+1$ scores. The event that its rank is at most k has probability $k/(n+1) \geq 1-\alpha$ and implies the future score is at most the calibration order statistic. Ties handled by the nonstrict comparison are conservative. The infinite-threshold case is immediate. □

This standard conformal principle is discussed in [5]. It is marginal over the exchangeable population, not per-pocket, per-horizon-subgroup or simultaneous path coverage. Chronological order alone does not establish exchangeability of overlapping time-series errors. Fitting $\widehat{s}$ or choosing pockets using the same calibration scores can also invalidate the stated construction.

10 Events, warnings and decision costs

Let τ be first entry into a declared event. Conditional hazards are $h_j = \mathbb{P}(\tau = j \mid \tau \geq j, \mathcal{F}_t)$. Chain probability gives

$$\mathbb{P}(\tau \leq P \mid \mathcal{F}_t) = 1 - \prod_{j=1}^{P} (1 - h_j). \tag{10.1}$$

No independence assumption is needed because hazards are conditional on survival. Replacing hazards by unconditional endpoint probabilities is invalid. Hazards $(0.1, 0.2, 0.15)$ give cumulative risks $(0.1, 0.28, 0.388)$.

For a single binary decision, let false-alarm cost be $c_{\mathrm{FA}} > 0$ and missed-event cost $c_{\mathrm{M}} > 0$, with no other costs. Warning has conditional expected cost $c_{\mathrm{FA}}(1-p)$ and silence has cost $c_{\mathrm{M}}p$, so warn when

$$p > \frac{c_{\mathrm{FA}}}{c_{\mathrm{FA}} + c_{\mathrm{M}}}. \tag{10.2}$$

Repeated alerts, intervention effects and uncertain calibration change this decision problem. A threshold derived for one decision is not a theorem for an entire adaptive monitoring policy.

11 A complete optimization worksheet

Suppose two eligible lens forecasts concern the same future scalar target, and a supplied error second-moment model is

$$M = \begin{pmatrix} 4 & 1 \\ 1 & 9 \end{pmatrix}. \tag{11.1}$$

Its determinant is $35 > 0$ and its first diagonal entry is positive, so M is positive definite. The inverse is $35^{-1}\left(\begin{smallmatrix} 9 & -1 \\ -1 & 4 \end{smallmatrix}\right)$. Multiplying by ones gives $35^{-1}(8,3)^{\top}$, with sum $11/35$. Thus weights are $(8/11, 3/11)$ and risk is $35/11$. If current forecasts are 10 and 14, the point forecast is $(80+42)/11 = 122/11$.

This is an optimal point combination under the supplied second moments. It does not specify a distribution for Y. To issue intervals, one may separately model matured residuals of that combination or specify a joint probabilistic model. Taking the spread between 10 and 14 as the entire forecast SD is unjustified: candidates can agree and both be wrong, or disagree despite small noise.

For a probability-mixture construction, suppose instead that the two objects are predictive laws with distinct mixture weights, $F = \frac{3}{4}N(0,1) + \frac{1}{4}N(2,4)$. Then

$$F(y) = \tfrac{3}{4}\Phi(y) + \tfrac{1}{4}\Phi((y-2)/2), \tag{11.2}$$

and probability in $[-1,1]$ is $\frac{3}{4}[\Phi(1)-\Phi(-1)] + \frac{1}{4}[\Phi(-1/2)-\Phi(-3/2)]$. This is a fully specified probability equation, conditional on the proposed normal components. A Gaussian with the same mean and variance is another model, not algebraically the same mixture.

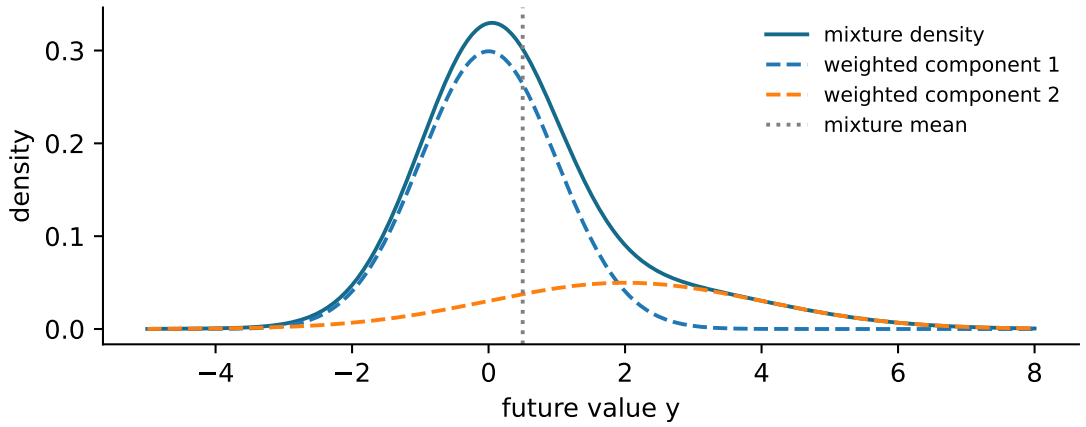


Figure 1: The specified two-component probability mixture and its weighted components. Its uncertainty contains both within-component spread and between-component disagreement. This is an analytical distribution, not a fitted forecast.

12 Scores for complete distributions and intervals

For a CDF F and observed y, the continuous ranked probability score is

$$\mathrm{CRPS}(F, y) = \int_{-\infty}^{\infty} [F(z) - \mathbf{1}\{y \le z\}]^2 \, dz. \tag{12.1}$$

Under finite first-moment conditions, its expected excess over the true CDF G is $\int (F - G)^2 dz$. To see this, fix z, apply the binary squared-loss decomposition to $\mathbf{1}\{Y \le z\}$, and integrate. Nonnegative integrands justify the expectation interchange; integrability ensures finite comparisons. This gives a mathematical reason to optimize the entire predictive distribution, not only one interval's coverage.

A central $(1 - \alpha)$ interval $[l, u]$ can be assessed using

$$\mathrm{IS}_\alpha(l, u; y) = (u - l) + \frac{2}{\alpha}(l - y)\mathbf{1}\{y < l\} + \frac{2}{\alpha}(y - u)\mathbf{1}\{y > u\}. \tag{12.2}$$

Width is penalized and misses incur an additional distance penalty. Coverage alone can be increased by arbitrarily wide intervals. Proper-score theory explains the role of such objectives [4]; their use does not establish a new calibration theorem for dependent WPM forecasts.

For integer targets, distinguish a continuous approximation from a discrete probability model. A rounded model with $Q = \mathrm{round}(Y)$ has mass $\mathbb{P}(Q = k) = F(k + 1/2) - F((k - 1/2)^-)$ under a stated boundary convention. Conditioning on an admissible set A gives $p_A(k) = p(k)\mathbf{1}\{k \in A\} / \sum_{j \in A} p(j)$ when the denominator is positive. A mathematical support constraint does not establish that the resulting probabilities are calibrated, and verifying an admissible candidate does not identify its sequence index.

13 Computational optimization versus predictive optimization

A numerical solver minimizes the chosen objective; it does not decide whether that objective predicts the intended target. Solver tolerance, candidate grammar, risk estimation and future-law assumptions are different uncertainty sources. A tiny numerical residual can coexist with a wrong model or an unidentified parameter.

For expensive landscapes, one may reuse sufficient summaries for a fixed dictionary, batch aligned probe operations or exploit sparse supports. These can accelerate arithmetic without changing its mathematical objective. Low-rank approximation or downsampling changes the representation unless an error bound or exact sufficiency result applies. Report computational approximation separately from predictive uncertainty.

If computational cost is itself part of the decision, an objective such as $R(r) + \lambda c(r)$ is legitimate when c and its units are declared. It optimizes a cost–risk trade-off, not pure predictive risk. Likewise, constrained optimization $\min R(r)$ subject to $c(r) \le B$ may select a different lens from unrestricted optimization. Cost weights and budgets are user or application choices, not universal WPM constants.

14 Application rules and solved exercises

1. Fix target, horizon, loss and causal cutoff. Separate location, scale, event and decision outputs.
2. Preserve Fixed as a legitimate policy; use Multi for justified same-target combinations and Auto only with a specified causal update rule.

3. Estimate error second moments from issued forecasts, retaining bias, dependence and group provenance.
4. Use constraints and regularization appropriate to the forecast meaning. A score-normalized weight is not a correctness probability.
5. Fit probability models and calibrators on separate permitted information. State the assumptions of every coverage claim.
6. Report event probabilities at the requested horizons and make actions using stated costs. Refresh forecasts only as observations actually arrive.

Exercise 14.1. For two candidates let $A = B = 1$ and $C = 0.8$. Find the optimum convex mixture and its risk.

Solution. $D = 0.4$ and $\alpha = 0.5$. Risk is $0.25 + 0.25 + 2(0.25)(0.8) = 0.9$. Strong dependence limits diversification.

Exercise 14.2. With $n = 19$ and nominal 90% split-conformal coverage, identify the order statistic.

Solution. $k = \lceil 20 \times 0.9 \rceil = 18$. Use the eighteenth smallest calibration score. The guarantee needs the stated exchangeability and frozen-model conditions.

Exercise 14.3. Let $c_{\mathrm{FA}} = 2$, $c_{\mathrm{M}} = 8$. At what probability is a single warning preferred?

Solution. The threshold is $2/(2 + 8) = 0.2$. At equality both actions have equal cost. This calculation assumes p is the relevant conditional event probability.

Scope of the mathematical exposition

Unless explicitly identified as a sourced evidence statement, numerical values are constructed arithmetic examples. Application requires specifying the observation process, fitting unknown quantities, and checking reliability for the intended population and horizon. The archival DOI for this study is 10.5281/zenodo.23782978.

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